



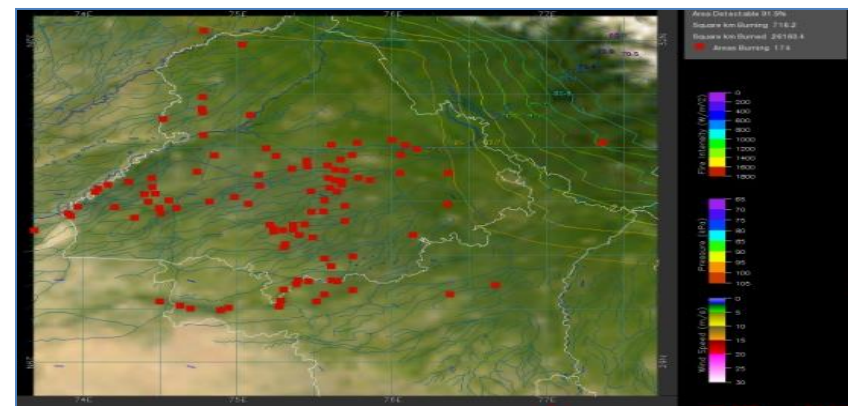
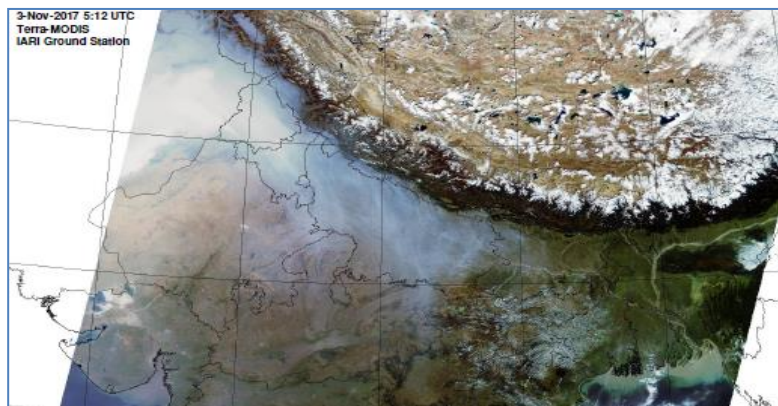
Bulletin No. 11

Events Date:
25-Sep-2023

Issued on:
25-Sep-2023



MONITORING PADDY RESIDUE BURNING IN INDIA USING SATELLITE REMOTE SENSING DURING 2023



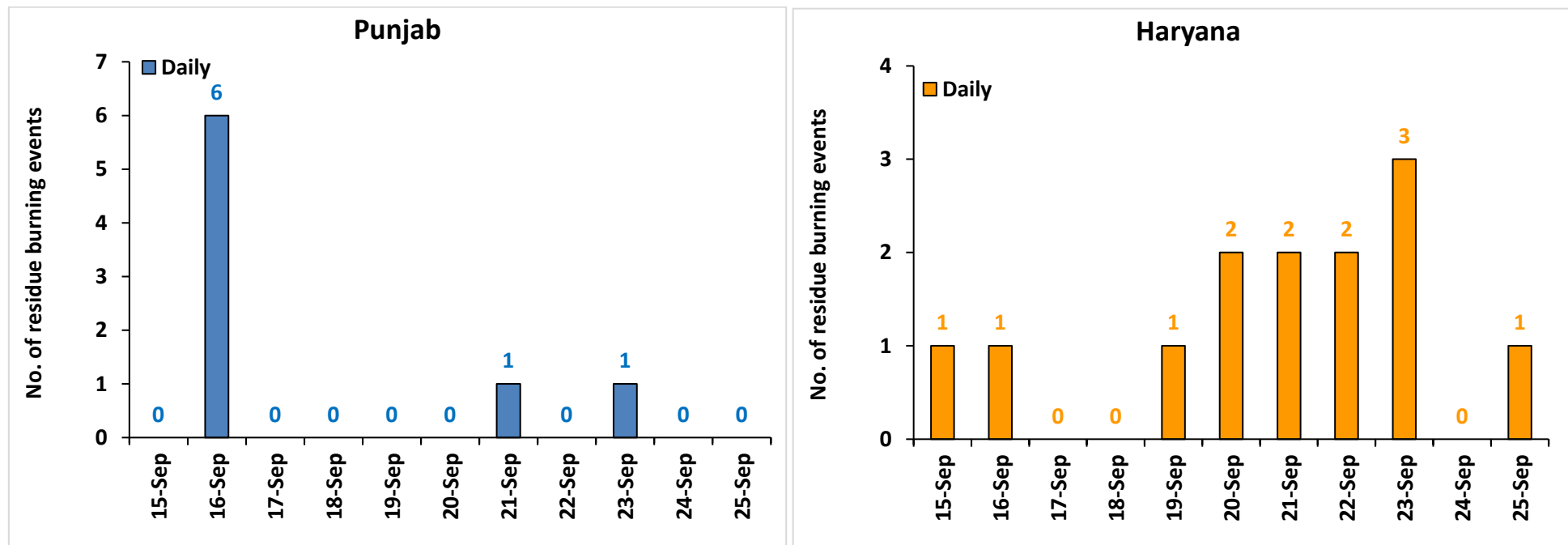
Consortium for Research on Agroecosystem Monitoring and Modeling from Space (CREAMS) Laboratory,
Division of Agricultural Physics, ICAR – Indian Agricultural Research Institute, New Delhi – 110012

<http://creams.iari.res.in>

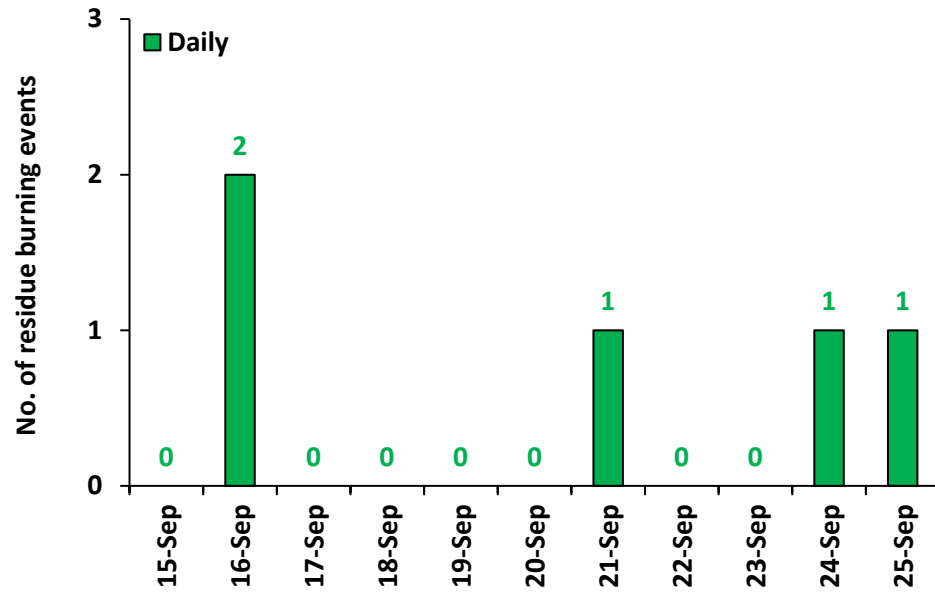
Highlights for 25-Sep-2023

- The active fire events due to rice residue burning were monitored using satellite remote sensing, following the "Standard Protocol for Estimation of Crop Residue Burning Fire Events using Satellite Data".
- Satellites detected **03** residue burning events in the six study States on 25-Sep-2023.
- The state wise events detected on 25-Sep-2023 were **00** in Punjab, **01** in Haryana, **01** in UP, **00** in Delhi, **00** in Rajasthan and **01** in MP.
- Total **29** burning events were detected in the six states between 15-Sept-2023 and 25-Sep-2023, which are distributed as **08, 13, 05, 00, 00** and **03** in Punjab, Haryana, UP, Delhi, Rajasthan and MP, respectively.
- Detection of residue burning events was hampered by the presence of clouds over Punjab, Delhi, Uttar Pradesh, Rajasthan and Madhya Pradesh states.

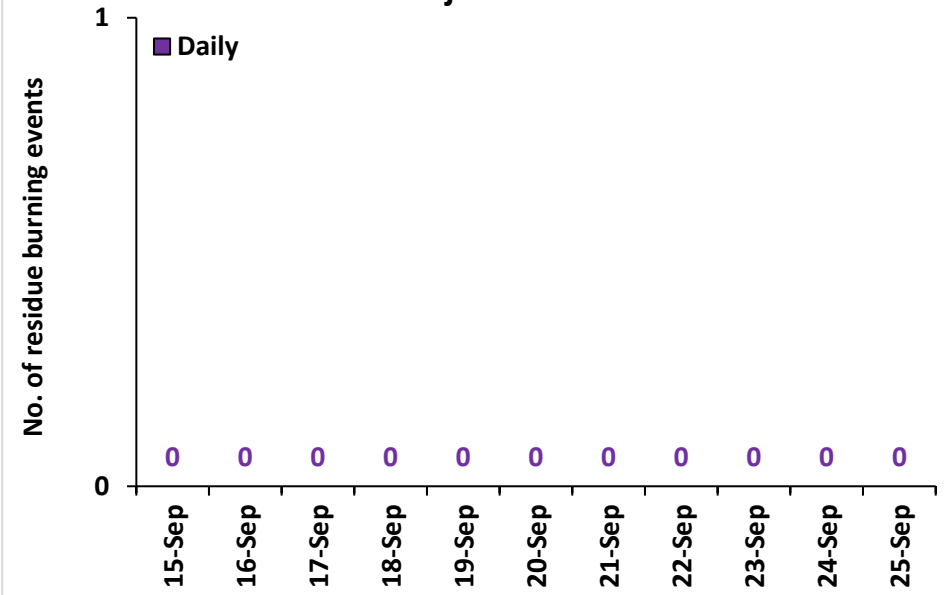
Temporal distribution of rice residue burning events for the study States in 2023



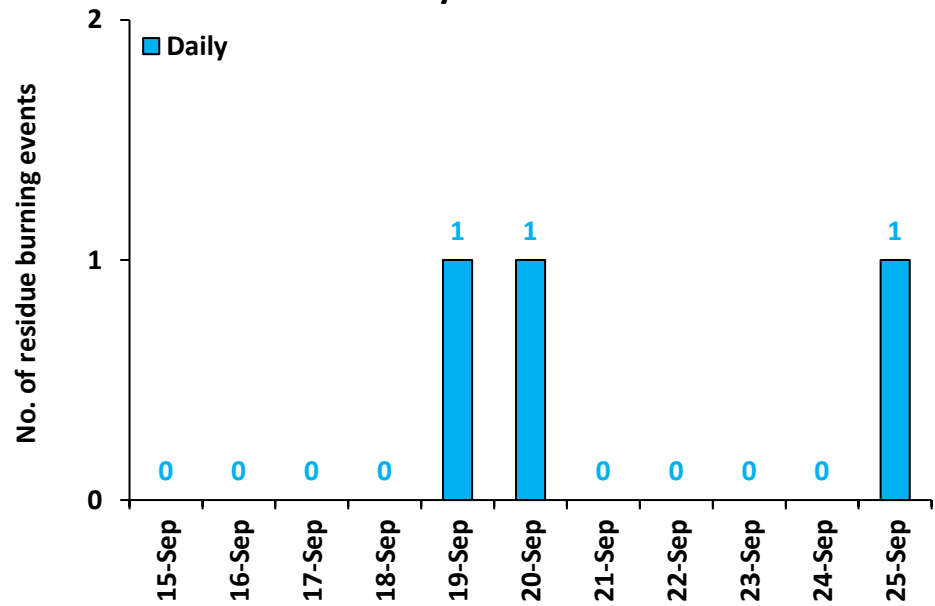
Uttar Pradesh



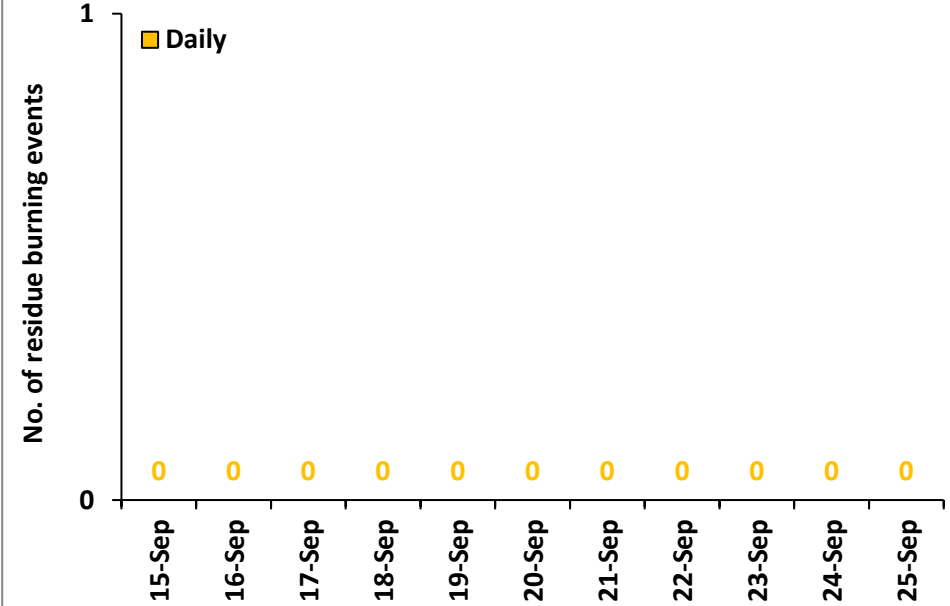
Rajasthan



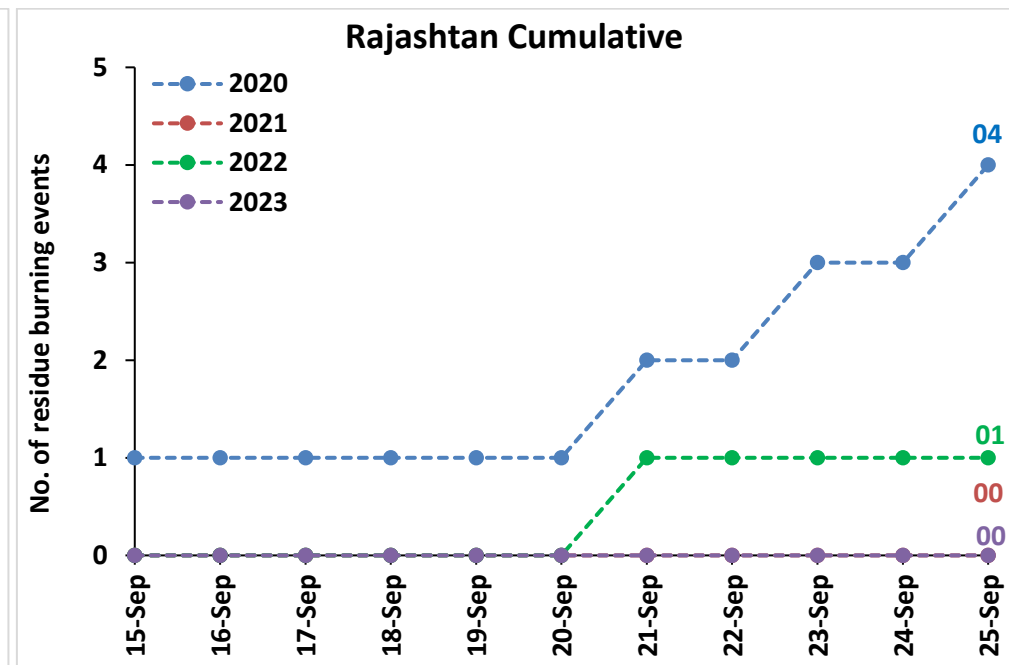
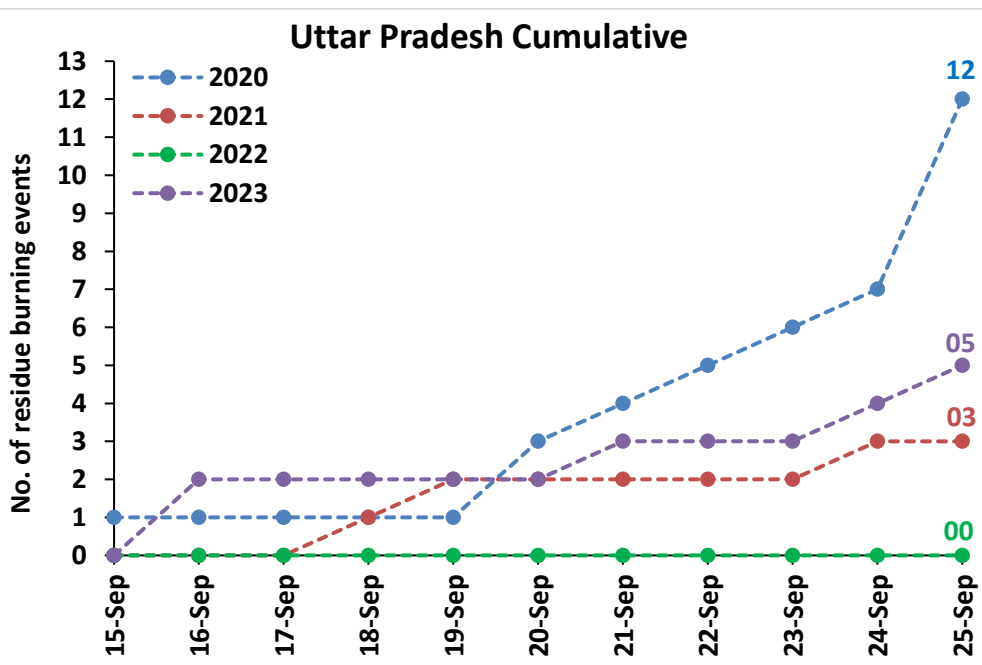
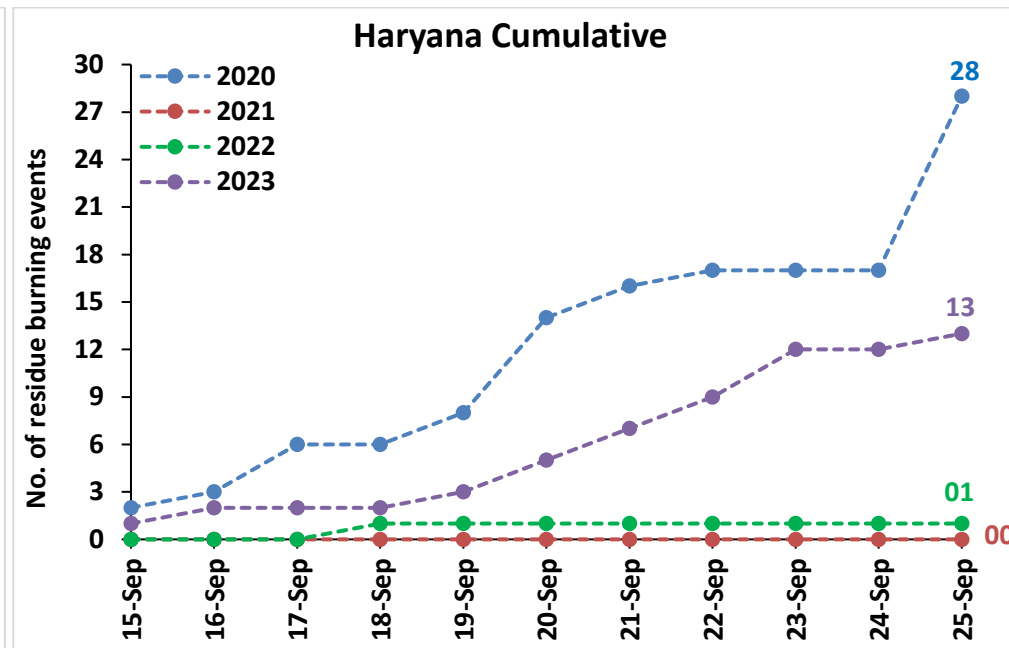
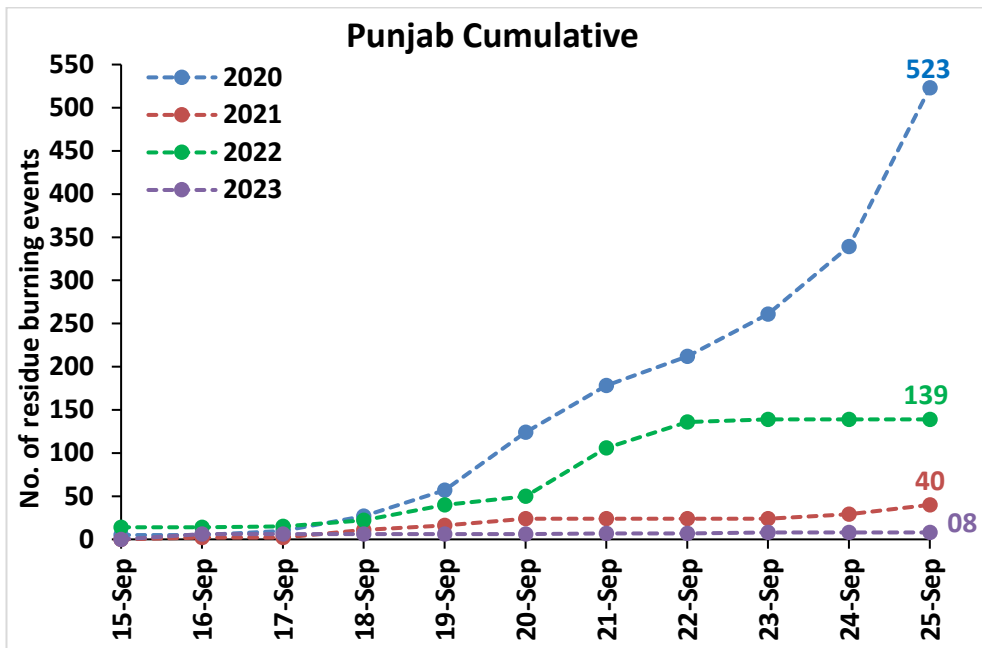
Madhya Pradesh

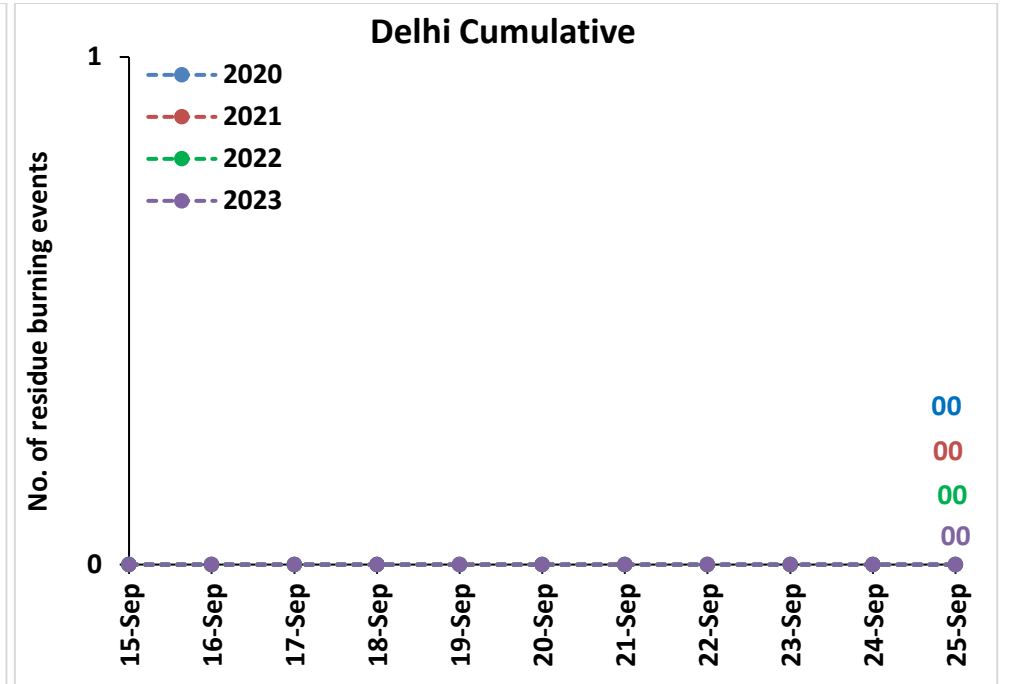
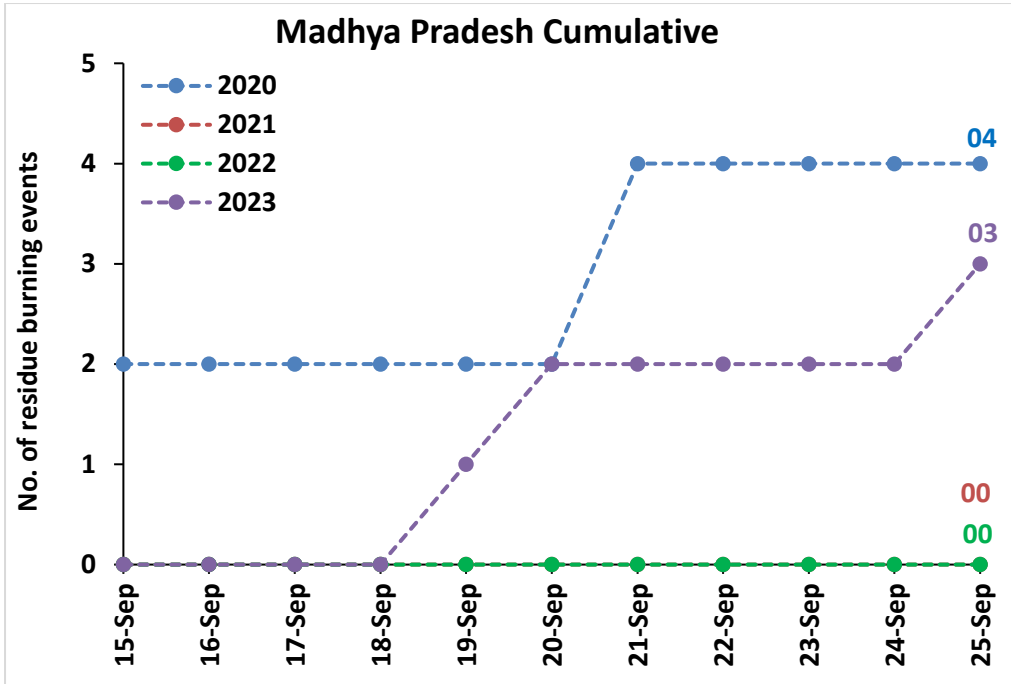


Delhi

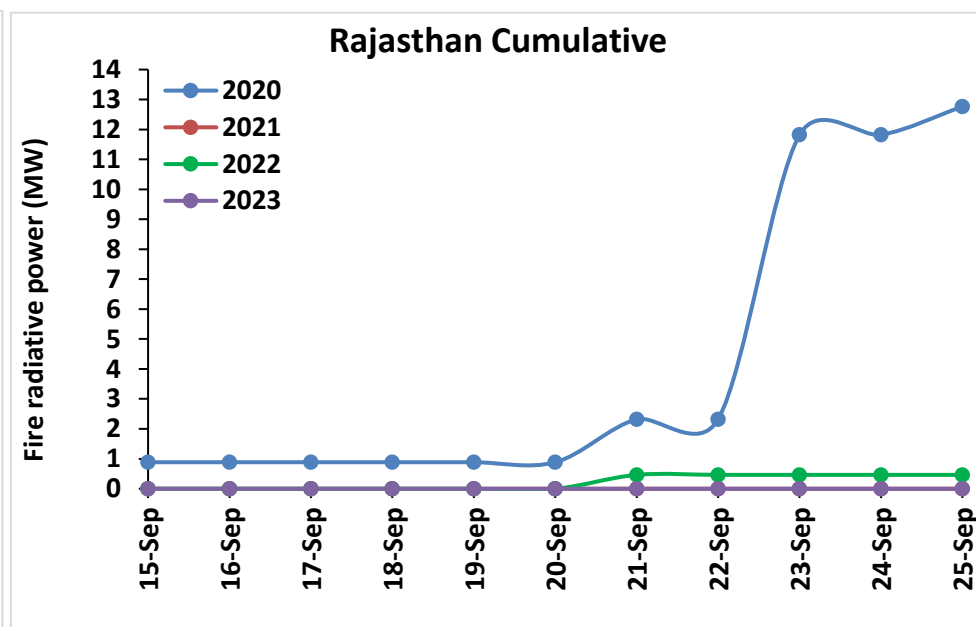
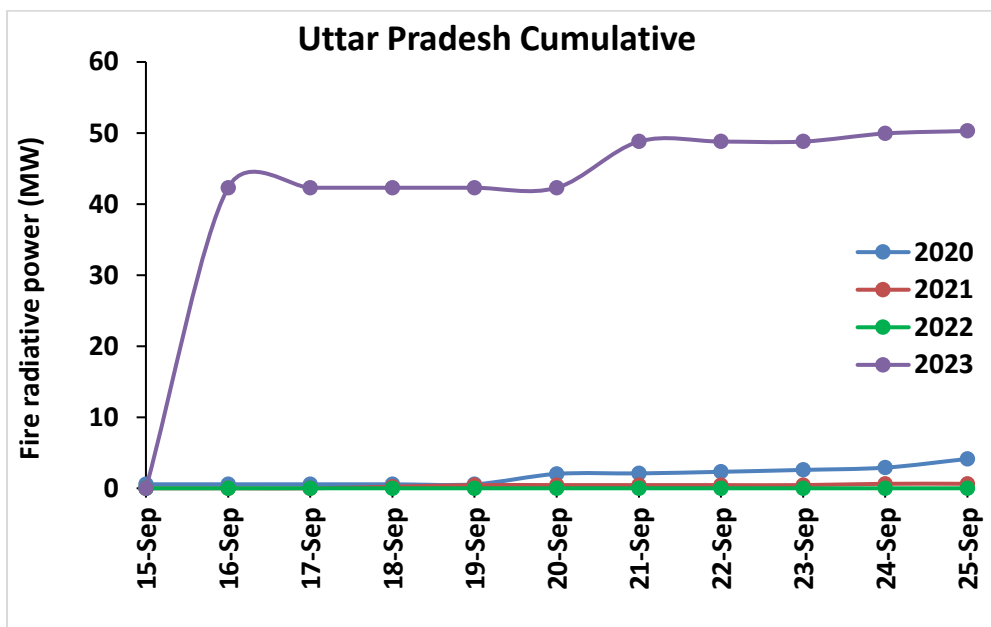
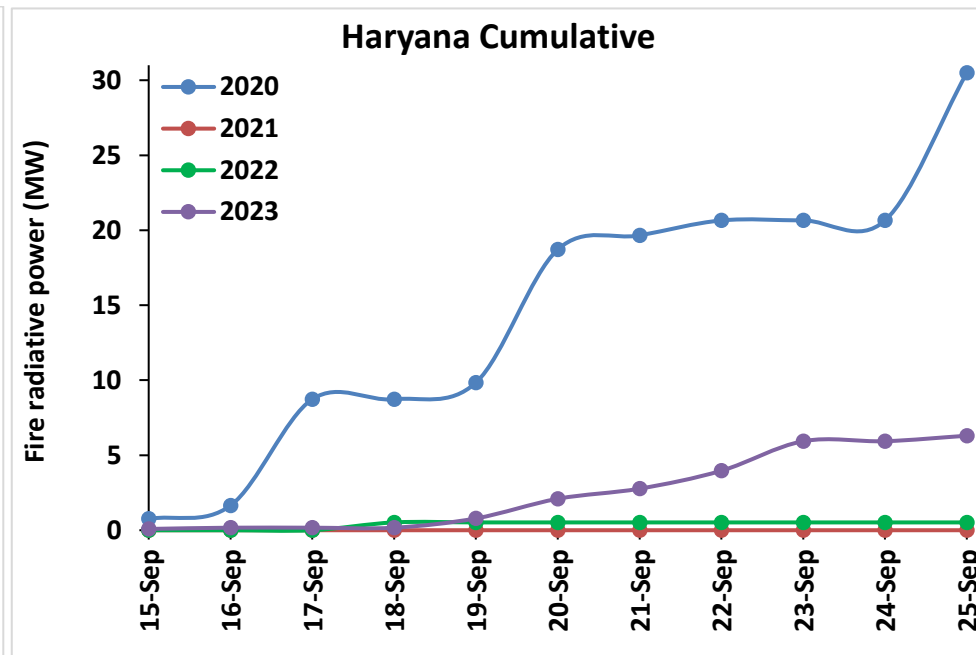
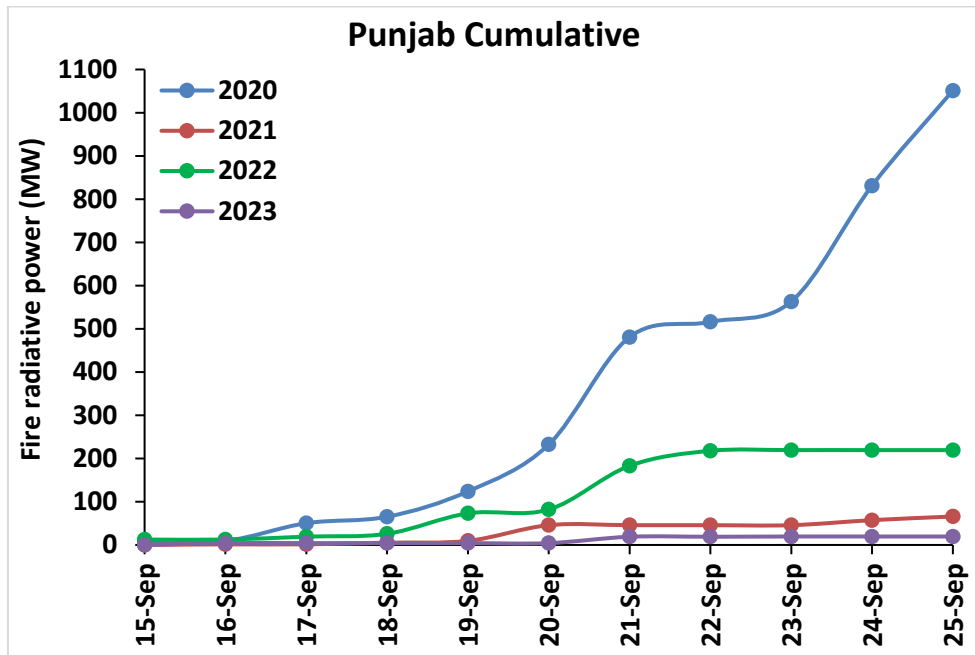


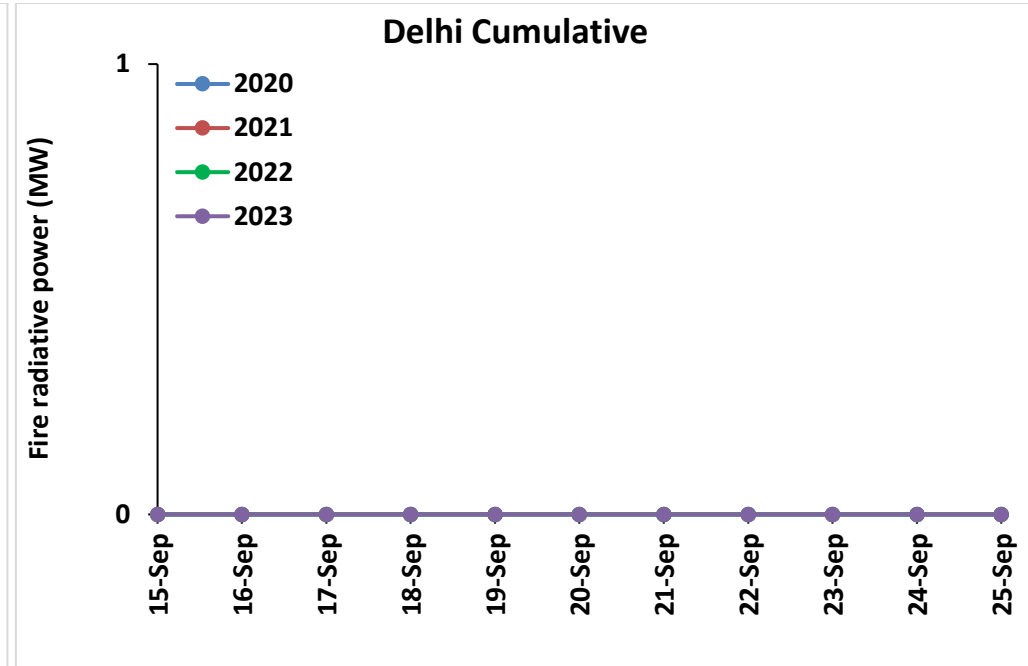
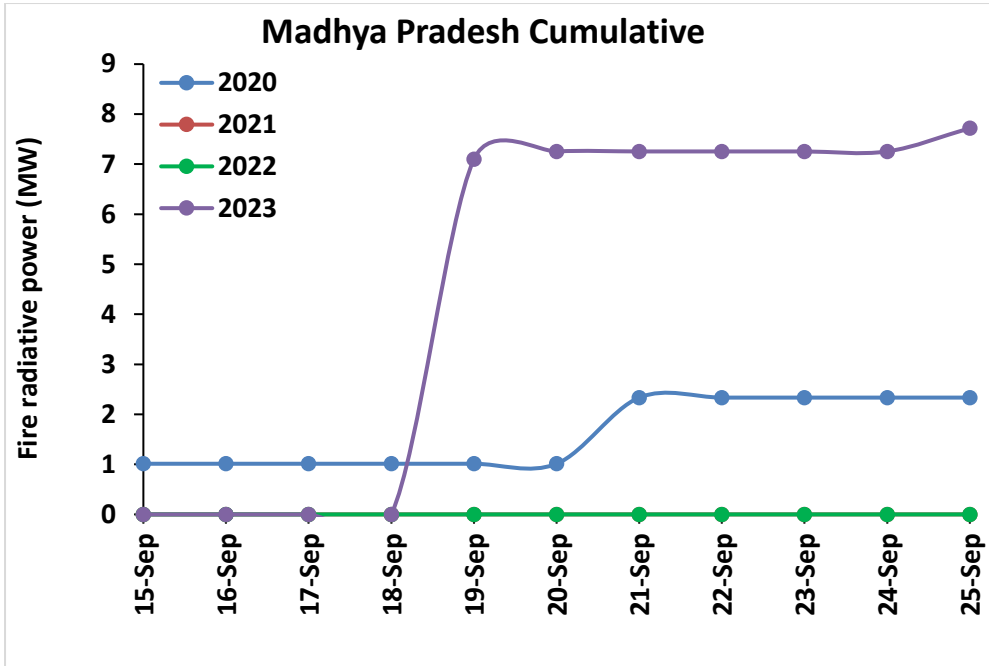
Comparison of residue burning events in current year (2023) with previous years (2020, 2021 and 2022) for the study State





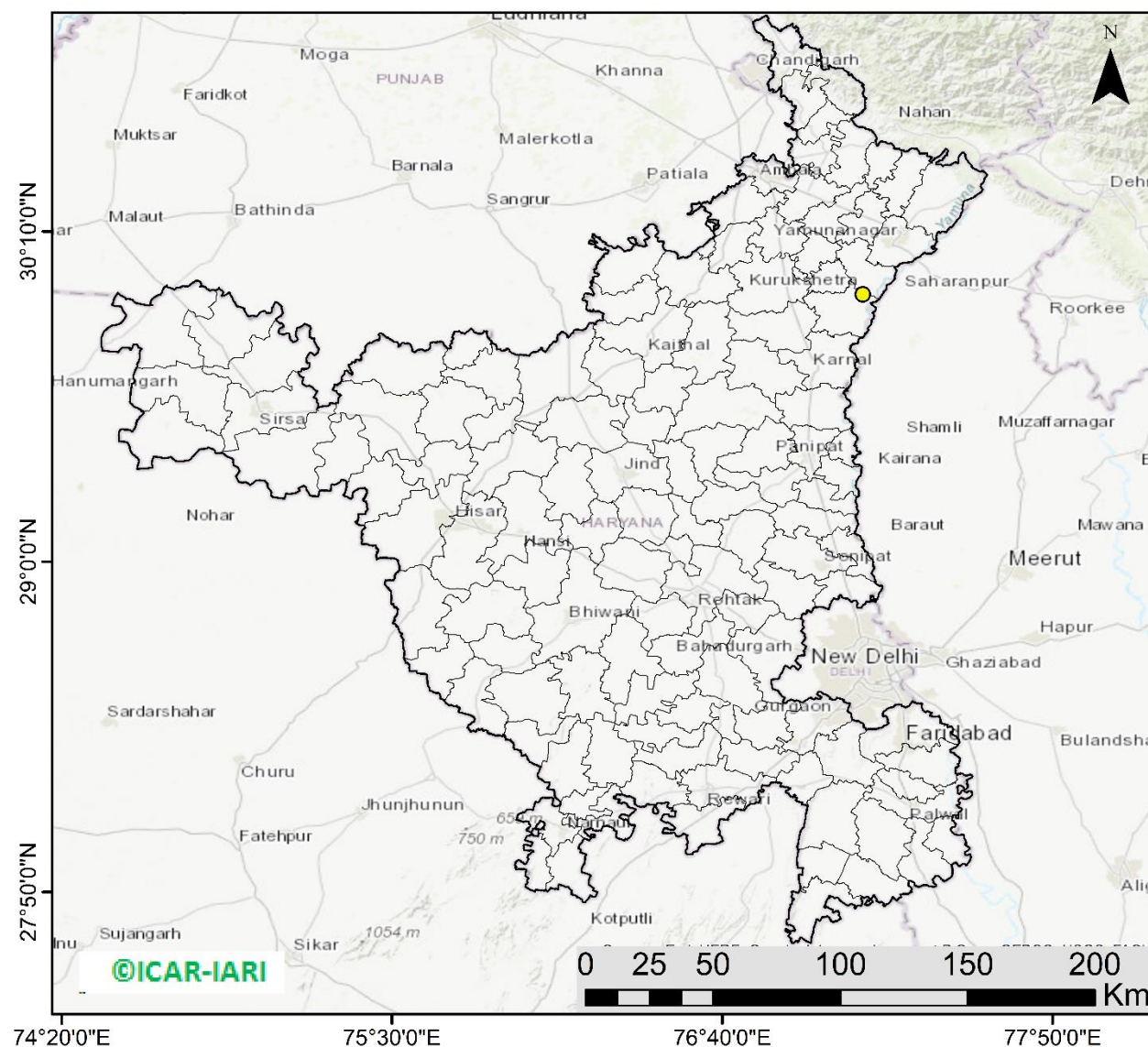
Comparison of residue burning fire power in current year (2023) with previous years (2020, 2021 and 2022) for the study States





Haryana

RICE RESIDUE BURNING IN HARYANA



01 burning events
detected in Haryana on
(25th Sep 2023)

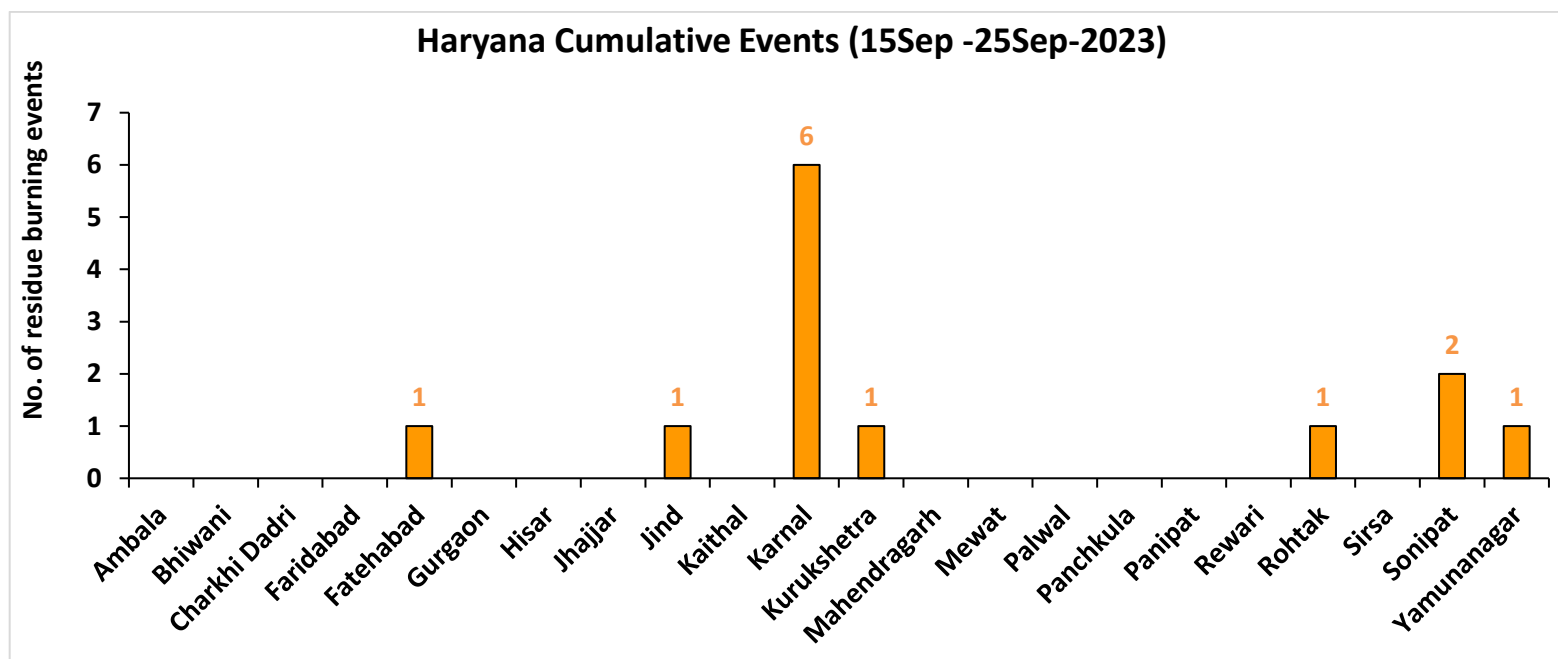
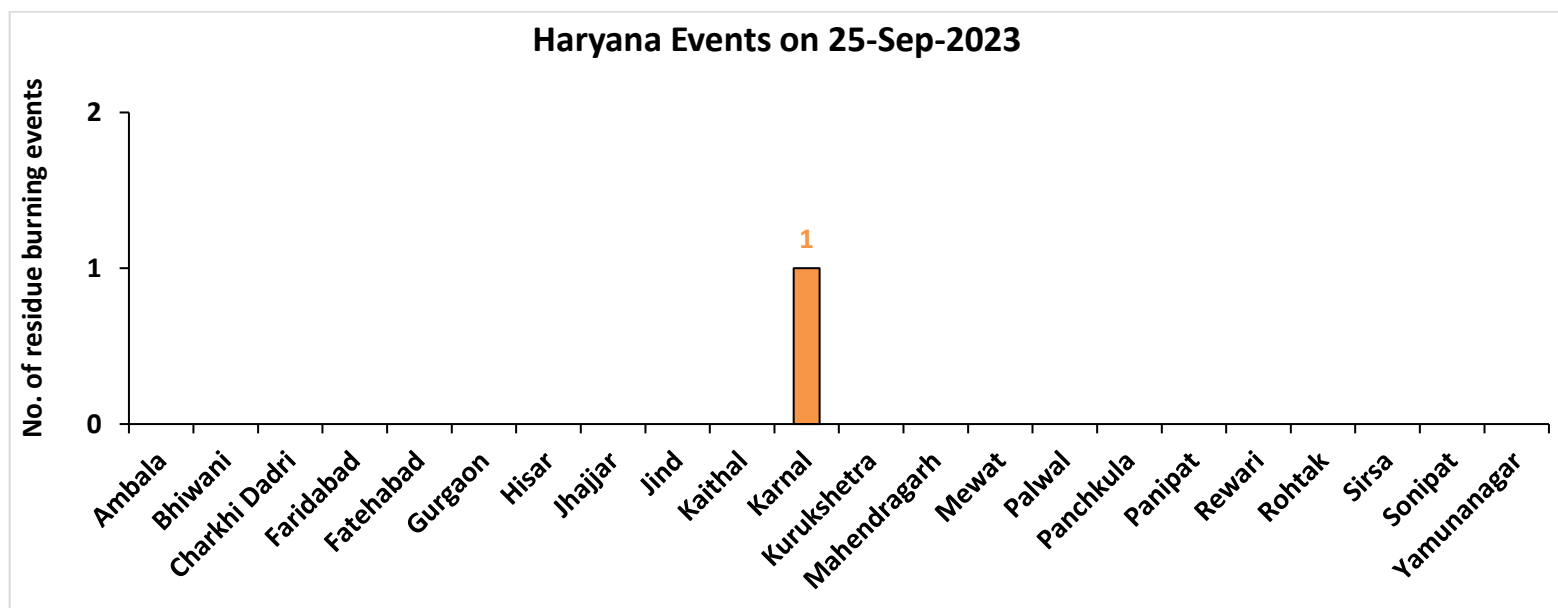
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



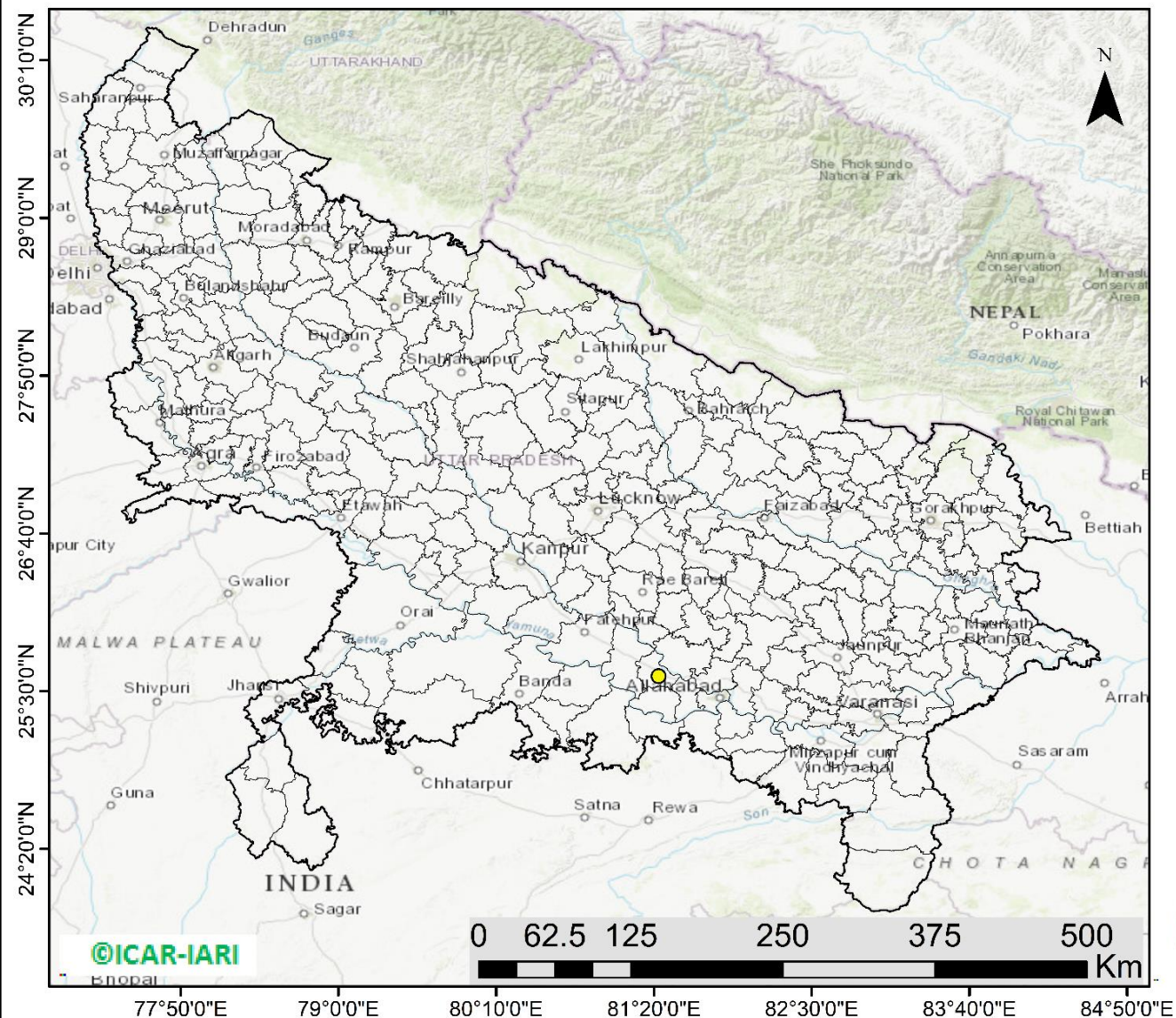
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Haryana (15Sep-25Sep-2023)



Uttar Pradesh

RICE RESIDUE BURNING IN UTTAR PRADESH



01 burning events
detected in Uttar Pradesh on
(25th Sep 2023)

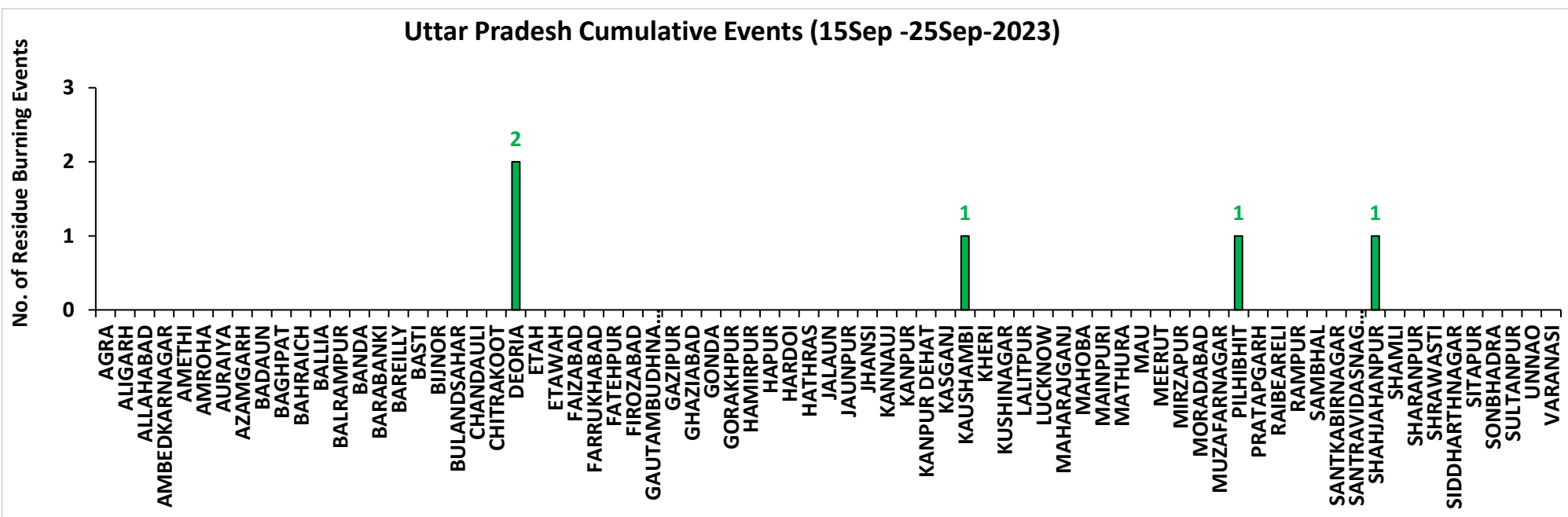
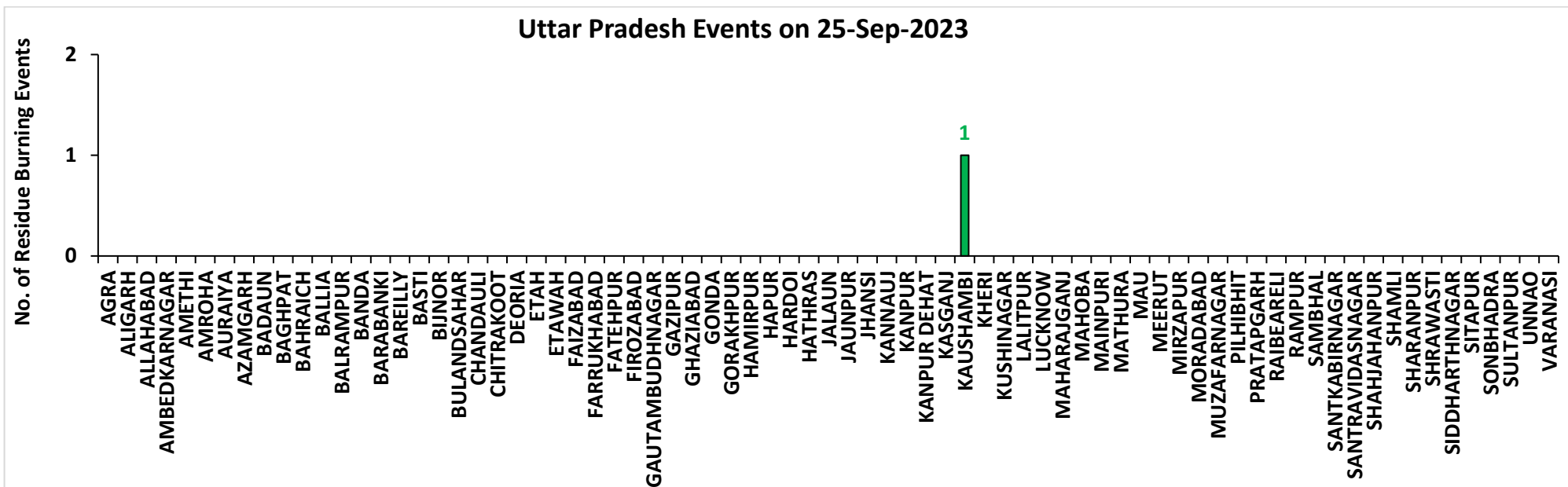
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



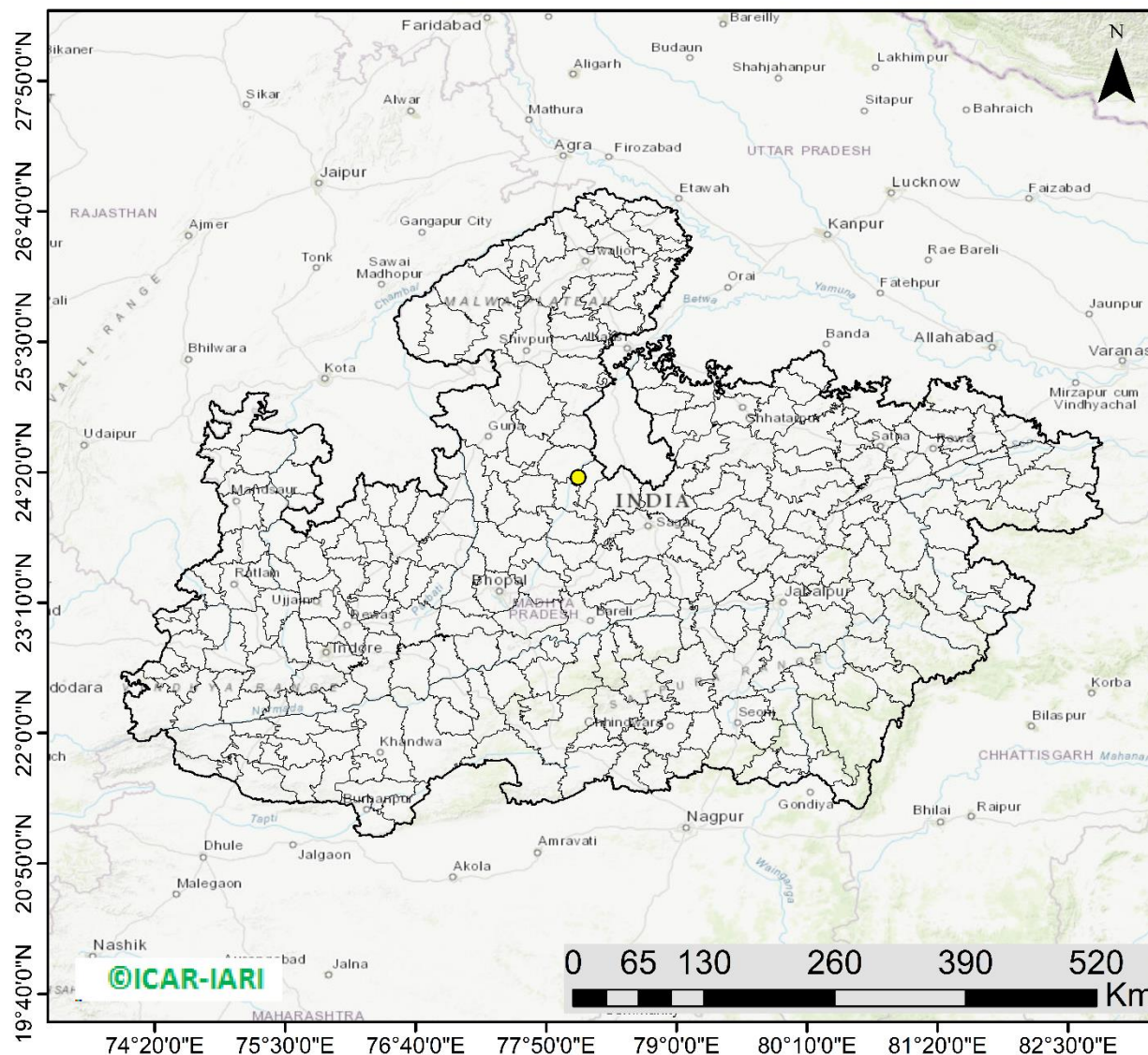
<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Uttar Pradesh (15Sep-25Sep-2023)



Madhya Pradesh

RICE RESIDUE BURNING IN MADHYA PRADESH



01 burning events
detected in MP on
(25th Sep 2023)

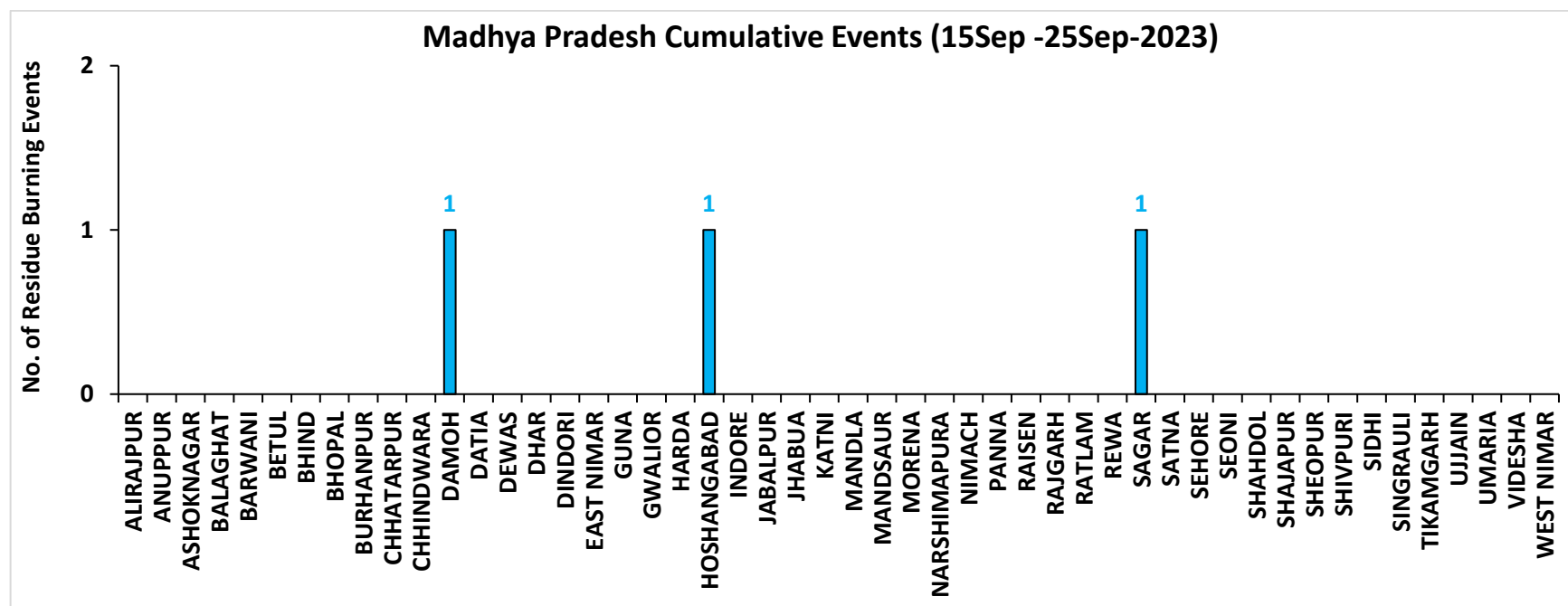
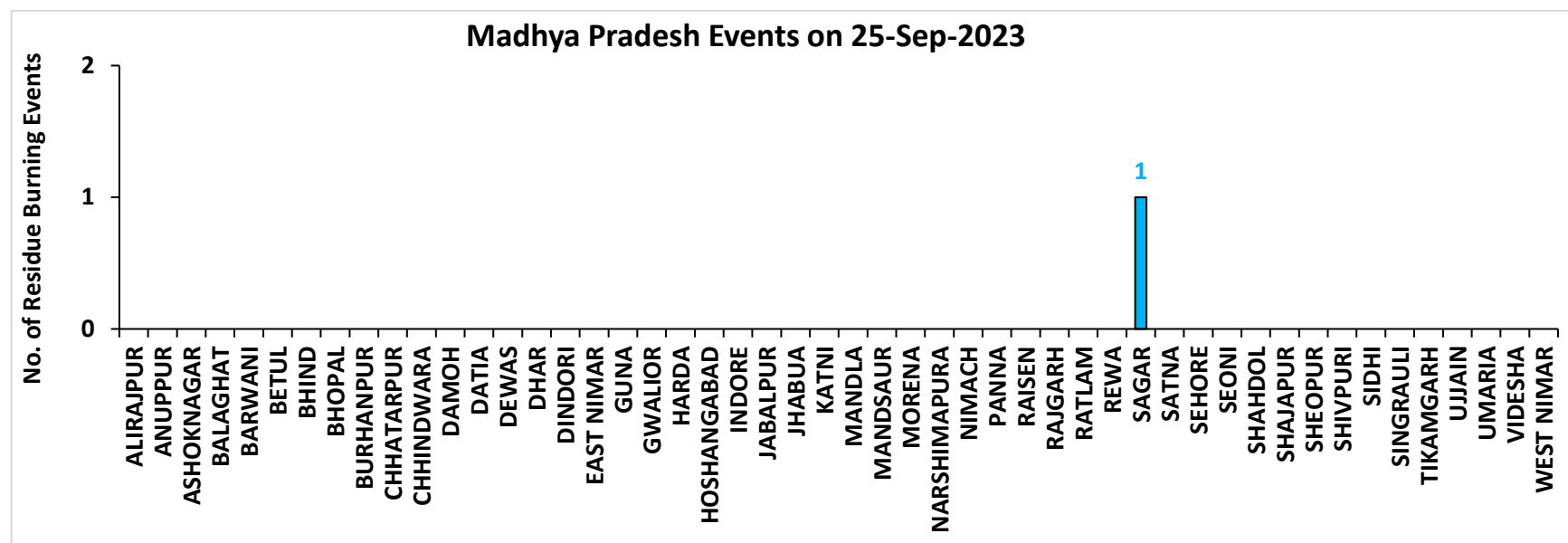
Fire Intensity (W/m²)

- 0 - 5
- 6 - 10
- 11 - 15
- 16 - 20
- >20



<http://geoportal.icar.gov.in:8080/geoexplorer/composer/>

District-wise cumulative number of residues burning events in Madhya Pradesh (15Sep-25Sep-2023)



District-wise cumulative number of residues burning events detected in 2020, 2021, 2022 and 2023
(Period: 15 September-25 September)

(a) Punjab

District	15 Sep-25 Sep			
	2020	2021	2022	2023
Amritsar	425	34	121	6
Barnala	1	0	0	0
Bathinda	0	0	0	0
Faridkot	1	1	0	0
Fatehgarh Sahib	0	0	0	0
Fazilka	0	0	0	0
Firozpur	2	0	0	1
Gurdaspur	5	0	1	0
Hoshiarpur	0	0	0	1
Jalandhar	2	0	0	0
Kapurthala	5	0	0	0
Ludhiana	5	1	0	0
Mansa	1	0	0	0
Moga	3	0	0	0
Muktsar	1	0	0	0
Pathankot	0	0	0	0
Patiala	11	3	1	0
Rupnagar	0	0	0	0
Sangrur	2	0	0	0
SAS NAGAR (Mohali)	3	0	1	0
SBS Nagar	0	0	1	0
Tarn Taran	56	1	14	0
Total	523	40	139	8

(b) Haryana

District	15 Sep-25 Sep			
	2020	2021	2022	2023
AMBALA	3	0	0	0
BHIWANI	0	0	0	0
CHARKHI DADRI	0	0	0	0
FARIDABAD	0	0	0	0
FATEHABAD	2	0	0	1
GURGAON	0	0	0	0
HISAR	0	0	0	0
JHAJJAR	0	0	0	0
JIND	0	0	1	1
KAITHAL	1	0	0	0
KARNAL	16	0	0	6
KURUKSHETRA	4	0	0	1
MAHENDRAGARH	0	0	0	0
MEWAT	0	0	0	0
PALWAL	0	0	0	0
PANCHKULA	0	0	0	0
PANIPAT	1	0	0	0
REWARI	0	0	0	0
ROHTAK	0	0	0	1
SIRSA	1	0	0	0
SONIPAT	0	0	0	2
YAMUNANAGAR	0	0	0	1
Total	28	0	1	13

(c) Uttar Pradesh

District	15 Sep-25 Sep												
	2020	2021	2022	2023	GAUTAMBUDHNAGAR	0	0	0	0	MUZAFARNAGAR	0	0	0
AGRA	0	0	0	0	GAZIPUR	0	0	0	0	PILHIBHIT	0	0	1
ALIGARH	1	0	0	0	GHAZIABAD	1	0	0	0	PRATAPGARH	0	0	0
ALLAHABAD	0	0	0	0	GONDA	0	0	0	0	RAIBEARELI	1	0	0
AMBEDKARNAGAR	0	0	0	0	GORAKHPUR	0	0	0	0	RAMPUR	0	1	0
AMETHI	0	0	0	0	HAMIRPUR	0	0	0	0	SAMBHAL	3	0	0
AMROHA	0	0	0	0	HAPUR	1	0	0	0	SANTKABIRNAGAR	0	0	0
AURAIYA	1	0	0	0	HARDOI	0	0	0	0	SANTRAVIDASNAGAR	0	0	0
AZAMGARH	0	0	0	0	HATHRAS	0	0	0	0	SHAHJAHANPUR	0	0	1
BADAUN	0	0	0	0	JALAUN	0	0	0	0	SHAMLI	1	0	0
BAGHPAT	1	0	0	0	JAUNPUR	0	0	0	0	SHARANPUR	0	0	0
BAHRAICH	0	0	0	0	JHANSI	0	0	0	0	SHRAWASTI	0	0	0
BALLIA	0	0	0	0	KANNAUJ	0	0	0	0	SIDDHARTHANAGAR	0	0	0
BALRAMPUR	0	0	0	0	KANPUR	0	1	0	0	SITAPUR	0	0	0
BANDA	0	0	0	0	KANPUR DEHAT	0	0	0	0	SONBHADRA	0	0	0
BARABANKI	0	0	0	0	KASGANJ	0	0	0	0	SULTANPUR	0	0	0
BAREILLY	0	0	0	0	KAUSHAMBI	0	0	0	1	UNNAO	0	0	0
BASTI	0	0	0	0	KHERI	0	0	0	0	VARANASI	0	0	0
BIJNOR	1	0	0	0	KUSHINAGAR	0	0	0	0	Total	12	3	5
BULANDSAHAR	0	0	0	0	LALITPUR	0	0	0	0				
CHANDAULI	0	0	0	0	LUCKNOW	0	0	0	0				
CHITRAKOOT	0	0	0	0	MAHARAJGANJ	0	0	0	0				
DEORIA	0	0	0	2	MAHOBAB	0	0	0	0				
ETAH	0	0	0	0	MAINPURI	0	0	0	0				
ETAWAH	0	0	0	0	MATHURA	0	1	0	0				
FAIZABAD	0	0	0	0	MAU	0	0	0	0				
FARRUKHABAD	0	0	0	0	MEERUT	1	0	0	0				
FATEHPUR	0	0	0	0	MIRZAPUR	0	0	0	0				
FIROZABAD	0	0	0	0	MORADABAD	0	0	0	0				

(d) Delhi

District	15 Sep-25 Sep			
	2020	2021	2022	2023
CENTRAL	0	0	0	0
EAST	0	0	0	0
NEW DELHI	0	0	0	0
NORTH	0	0	0	0
NORTH EAST	0	0	0	0
NORTH WEST DELHI	0	0	0	0
SHAHADRA	0	0	0	0
SOUTH	0	0	0	0
SOUTH EAST	0	0	0	0
SOUTH WEST DELHI	0	0	0	0
WEST DELHI	0	0	0	0
Total	0	0	0	0

(e) Rajasthan

District	15 Sep-25 Sep			
	2020	2021	2022	2023
AJMER	0	0	0	0
ALWAR	0	0	0	0
BANSWARA	0	0	0	0
BARAN	0	0	0	0
BARMER	1	0	0	0
BHARATPUR	0	0	0	0
BHILWARA	1	0	0	0
BIKANER	0	0	0	0
BUNDI	0	0	1	0
CHITTAURGARH	0	0	0	0
CHURU	0	0	0	0
DAUSA	0	0	0	0
DHOLPUR	0	0	0	0
DUNGARPUR	0	0	0	0
GANGANAGAR	0	0	0	0
HANUMANGARH	0	0	0	0
JAIPUR	0	0	0	0
JAISELMER	1	0	0	0
JALOR	0	0	0	0
JHALAWAR	0	0	0	0
JHUNJHUNUN	0	0	0	0
JODHPUR	1	0	0	0
KARALI	0	0	0	0
KOTA	0	0	0	0
NAGPUR	0	0	0	0
PALI	0	0	0	0
PRATAPGARH	0	0	0	0
RAJSMAND	0	0	0	0
SIKAR	0	0	0	0
SIROHI	0	0	0	0
SWAIMADHOPUR	0	0	0	0
TONK	0	0	0	0
UDAIPUR	0	0	0	0
Total	4	0	1	0

(f) Madhya Pradesh

District	15 Sep-25 Sep								
	2020	2021	2022	2023	PANNA	0	0	0	0
ALIRAJPUR	0	0	0	0	RAISEN	0	0	0	0
ANUPPUR	0	0	0	0	RAJGARH	0	0	0	0
ASHOKNAGAR	0	0	0	0	RATLAM	0	0	0	0
BALAGHAT	0	0	0	0	REWA	0	0	0	0
BARWANI	0	0	0	0	SAGAR	2	0	0	1
BETUL	0	0	0	0	SATNA	0	0	0	0
BHIND	0	0	0	0	SEHORE	0	0	0	0
BHOPAL	0	0	0	0	SEONI	0	0	0	0
BURHANPUR	0	0	0	0	SHAHDOL	0	0	0	0
CHHATARPUR	0	0	0	0	SHAJAPUR	0	0	0	0
CHHINDWARA	0	0	0	0	SHEOPUR	0	0	0	0
DAMOH	0	0	0	1	SHIVPURI	0	0	0	0
DATIA	0	0	0	0	SIDHI	0	0	0	0
DEWAS	0	0	0	0	SINGRAULI	0	0	0	0
DHAR	0	0	0	0	TIKAMGARH	0	0	0	0
DINDORI	0	0	0	0	UJJAIN	0	0	0	0
EAST NIMAR	0	0	0	0	UMARIA	0	0	0	0
GUNA	0	0	0	0	VIDESHA	0	0	0	0
GWALIOR	0	0	0	0	WEST NIMAR	0	0	0	0
HARDA	2	0	0	0	Total	4	0	0	3
HOSHANGABAD	0	0	0	1					
INDORE	0	0	0	0					
JABALPUR	0	0	0	0					
JHABUA	0	0	0	0					
KATNI	0	0	0	0					
MANDLA	0	0	0	0					
MANDSAUR	0	0	0	0					
MORENA	0	0	0	0					
NARSHIMAPURA	0	0	0	0					
NIMACH	0	0	0	0					

State-wise and date-wise number of paddy residue burning events detected in 2020 ,2021 ,2022 and 2023
(Period: 15 September-25 September)

Number of Paddy Residue Burning Events Detected												
Punjab					Haryana				Uttar Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	5	0	14	0	2	0	0	1	1	0	0	0
16-Sep	0	2	0	6	1	0	0	1	0	0	0	2
17-Sep	5	0	1	0	3	0	0	0	0	0	0	0
18-Sep	17	9	7	0	0	0	1	0	0	1	0	0
19-Sep	30	5	18	0	2	0	0	1	0	1	0	0
20-Sep	67	8	10	0	6	0	0	2	2	0	0	0
21-Sep	54	0	56	1	2	0	0	2	1	0	0	1
22-Sep	34	0	30	0	1	0	0	2	1	0	0	0
23-Sep	49	0	3	1	0	0	0	3	1	0	0	0
24-Sep	78	5	0	0	0	0	0	0	1	1	0	1
25-Sep	184	11	0	0	11	0	0	1	5	0	0	1
Total	523	40	139	8	28	0	1	13	12	3	0	5

Number of Paddy Residue Burning Events Detected

Delhi					Rajasthan				Madhya Pradesh			
Date	2020	2021	2022	2023	2020	2021	2022	2023	2020	2021	2022	2023
15-Sep	0	0	0	0	1	0	0	0	2	0	0	0
16-Sep	0	0	0	0	0	0	0	0	0	0	0	0
17-Sep	0	0	0	0	0	0	0	0	0	0	0	0
18-Sep	0	0	0	0	0	0	0	0	0	0	0	0
19-Sep	0	0	0	0	0	0	0	0	0	0	0	1
20-Sep	0	0	0	0	0	0	0	0	0	0	0	1
21-Sep	0	0	0	0	1	0	1	0	2	0	0	0
22-Sep	0	0	0	0	0	0	0	0	0	0	0	0
23-Sep	0	0	0	0	1	0	0	0	0	0	0	0
24-Sep	0	0	0	0	0	0	0	0	0	0	0	0
25-Sep	0	0	0	0	1	0	0	0	0	0	0	1
Total	0	0	0	0	4	0	1	0	4	0	0	3

Details of residue burning events in Haryana on 25-Sep-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	KARNAL	INDRI	S-NPP	29.94354	77.16318	13:10:54	D	2.60

Details of residue burning events in Uttar Pradesh on 25-Sep-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	KAUSHAMBI	SIRATHU	S-NPP	25.61001	81.36871	13:10:54	D	2.50

Details of residue burning events in Madhya Pradesh on 25-Sep-2023

S. No.	District	Tehsil / Block	Satellite	Latitude	Longitude	ACQ_TIME	Day / Night	Fire Power (W/m2)
1	SAGAR	BINA	S-NPP	24.28566	78.12108	13:10:54	D	3.30

The study used data received from following Satellites at NRSC Ground Station

S. No.	Satellite Name	Sensor	Resolution (meter)	Day / Night Passes
1.	Suomi NPP	VIIRS	375 / 750	Both
2.	Terra	MODIS	1000	Both
3.	Aqua	MODIS	1000	Both

**The rice mask for the Punjab and Haryana were provided by MNCFC, MoA&FW, Govt of India
The kharif crop mask for Delhi, Uttar Pradesh, Rajasthan and Madhya Pradesh were provided by NRSC (ISRO)**

Study Team

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GIS Maps of fire events can be visualized online on ICAR KRISHI Geoportal website: <http://geoportal.icar.gov.in:8080/geoexplorer/composer/>
(Part of KRISHI Portal: <https://krishi.icar.gov.in> initiative)
